

AWS E-Commerce Profit Leakage Analytics

Professional End-to-End Cloud Analytics Project Documentation

Project Overview

This project is an end-to-end cloud analytics solution built using AWS services and Power BI to analyze e-commerce sales, profitability, customer behavior, and shipping performance. The workflow demonstrates practical cloud analytics implementation including data ingestion, transformation, querying, and dashboard visualization.

Project Objectives

- Build a complete cloud analytics pipeline using AWS services.
- Identify profit leakage and business inefficiencies.
- Perform SQL-based business analysis using Amazon Athena.
- Create interactive Power BI dashboards for executive reporting.
- Demonstrate industry-level data analytics workflow for portfolio and job applications.

Technology Stack

Amazon S3 – Cloud storage for raw dataset

AWS Glue DataBrew – Data cleaning and preparation

AWS Glue – ETL workflow and data catalog creation

Amazon Athena – SQL querying and business analysis

Power BI – Interactive dashboard visualization

SQL – KPI calculations and analytics

Dataset Information

The project uses the Superstore sales dataset containing sales transactions, customer details, product categories, shipping information, discounts, regional performance, and profitability metrics.

Architecture Workflow

1. Upload raw dataset to Amazon S3.
2. Clean and transform data using AWS Glue DataBrew.
3. Perform ETL operations with AWS Glue.

4. Create and manage metadata using AWS Glue Catalog.
5. Execute SQL queries using Amazon Athena.
6. Connect processed data to Power BI.
7. Build interactive business dashboards.

Dashboard Pages

Page 1 – Executive Overview

Sales KPIs, profit trends, and regional performance.

Page 2 – Profit Leakage Analysis

Loss-making products, discount impact, and profit margin analysis.

Page 3 – Customer Analytics

Customer segmentation, top customers, and sales distribution.

Page 4 – Shipping Analytics

Shipping performance, delivery profitability, and order trends.

Key Business Insights

- High discounts significantly reduce profitability.
- Furniture category experiences major profit leakage.
- Consumer segment contributes the highest sales volume.
- Technology category generates stronger profit margins.
- Standard shipping mode dominates overall order volume.

Challenges Faced

- Handling null and inconsistent records.
- Configuring AWS Glue workflows and Athena queries.
- Maintaining consistent dashboard themes and layouts.
- Designing professional Power BI dashboards suitable for portfolio presentation.

Skills Demonstrated

- Cloud Analytics using AWS
- ETL Workflow Design
- SQL Querying with Athena

- Power BI Dashboard Development
- Data Cleaning and Transformation
- Business Intelligence and KPI Analysis

Conclusion

This project demonstrates a complete modern cloud analytics workflow using AWS services and Power BI. It showcases practical business intelligence implementation and highlights strong analytical, visualization, and cloud data processing capabilities suitable for data analyst and cloud analytics roles.

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